

## PKCERT AI & Software Development Internship

### Task 11 – Comparative Analysis of Classification Models

Objective:

Develop practical skills in building, evaluating, and comparing Logistic Regression, Random Forest, and Support Vector Machine (SVM) models using a common dataset.

Total Marks: 100

#### Part A – Dataset Selection & Preparation (15 Marks)

1. Select a public classification dataset.
2. Describe the dataset, features, and target variable.
3. Preprocess the dataset and split it into training and testing sets.

#### Part B – Model Development (45 Marks)

1. Build a Logistic Regression model.
2. Build a Random Forest model.
3. Build an SVM model.
4. Train and test each model using the same dataset.

#### Part C – Model Evaluation & Comparison (30 Marks)

1. Evaluate each model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
2. Prepare a comparison table summarizing the performance metrics.
3. Discuss the strengths and weaknesses of each model.

#### Part D – Recommendation & Conclusion (10 Marks)

1. Identify the best-performing model and justify your conclusion using the evaluation results.

| Assessment Area               | Marks |
|-------------------------------|-------|
| Dataset Preparation           | 15    |
| Model Development             | 45    |
| Model Evaluation & Comparison | 30    |
| Recommendation & Conclusion   | 10    |

Total

100